To create a snapshot training set of the streaming data using the Databricks Feature Engineering API client, we need the data from *Chapter 5*. To double-check that you have all the data necessary to complete this chapter's project work, you can use the Catalog view. Navigate to the `ml_in_action` catalog, then click on the `synthetic_transactions` schema, and confirm that you see the following tables: `product_3minute_max_price_ft`, `raw_transactions`, `transaction_count_history` and function `product_difference_ratio_on_demand_feature`. If you have all the necessary tables, you can skip the following steps and start with the *Chapter 6* notebook. Otherwise, please follow these steps first to stream in the Transactions data:

1. Start the multiple CPU cluster from *Chapter 5*.
2. Open the *Chapter 5* notebooks. Feel free to edit `CH5-01` to a smaller number of total steps by reducing the value of the `total` variable in the final cell of the notebook.
3. Hit **Run All** at the top of `CH5-01`. Confirm that the final cell titled "Loop for Generating Data" has begun to execute before moving on to the next step.
4. Hit **Run All** at the top of `CH5-02`. Confirm that the final cell title "readStream" has begun to execute before moving on to the next step.
5. Hit **Run All** at the top of `CH5-03`. Watch to see that the final two cells titled "Read and Write Streams" and "Writing the history of transaction counts to a table" have begun to execute before moving on to the next step.
6. While you are waiting for the data streams to finish, add a monitor to the `raw_transactions` table, as shown in previous chapters. You can refer back to the *Monitoring data quality with Databricks Lakehouse Monitoring* section in *Chapter 4*.
7. Once the generating notebook, `CH5-01`, finishes, you can sequentially terminate notebooks `CH5-02` and `CH5-03`.
8. Hit **Run All** at the top of notebook `CH5-04`.

Once all the Streaming Transactions tables are ready, you can begin this chapter's project work. To follow along in your own workspace, please refer to the following notebook:

- `CH6-01-Building Synthetic Transactions Training Set`

We begin by creating a training set, which we will build into a snapshot of data for training. To do so, we specify the feature lookups for the base DataFrame. In our case, the base DataFrame is the `raw_transactions` table. However, we cannot train on data without feature values present in the right tables. Data is continuously flowing into the `raw_transactions` table. Therefore, we need the base DataFrame to contain only transactions that have a calculated feature value. For that, we take the maximum of the two minimum timestamps. This is shown in the notebook code.

Figure 6.16 shows the feature objects we will use to build the training dataset. Note that for this project we're using Feature Lookups as well as a Feature Function. First, we include our feature tables `transaction_count_history` and `product_3min_max_price_ft`. Then we add the feature function `product_difference_ratio_on_demand_feature`.